

A WKB-based numerical method for capturing trait concentration in a dispersal evolution model

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Abstract

The evolution of a dispersal trait is a classical question in evolutionary ecology. A typical phenomenon is that, in the rare mutation regime, the trait distribution concentrates toward a Dirac mass, which makes the accurate numerical capture of the fittest trait difficult. In this paper, we consider a WKB reformulation of a dispersal evolution model and propose a semi-discrete numerical method for resolving the concentration in the trait direction without using prohibitively fine trait grids. We present the detailed semi-discrete scheme, establish positivity of the amplitude variable, and derive a fixed-grid stationary AP structure under an explicit one-sided stability condition on the weighted WKB remainder. We also identify a density-compatible WKB variant for which this remainder condition follows from discrete summation by parts. These results clarify both the useful structure and the current limitations of the method. Numerical experiments are organized to compare the proposed WKB formulation with direct discretizations, to monitor the weighted remainder, and to illustrate its advantage in locating the fittest trait in the small-mutation regime.

Key words. dispersal evolution, trait concentration, WKB ansatz, asymptotic-preserving method, Hamilton–Jacobi equation, principal eigenvalue.

1 Introduction

Trait concentration in the small-mutation regime is a major source of difficulty in the numerical approximation of space–trait structured population models. When the mutation parameter is small, the solution typically develops a sharp peak in the trait variable, while the dominant trait is selected through a delicate interaction between spatial heterogeneity, dispersal, and competition. In this regime, a direct discretization of the original density often requires a prohibitively fine mesh in the trait direction in order to resolve the narrow peak and identify the fittest trait accurately. As a consequence, the main numerical issue is not merely to approximate the full density, but to capture the location and structure of the emerging concentration efficiently and robustly.

In this work, we consider a representative dispersal evolution model of the form

$$\varepsilon \partial_t n_\varepsilon - D(\theta) \Delta_{\mathbf{x}} n_\varepsilon - \varepsilon^2 \Delta_\theta n_\varepsilon = n_\varepsilon (K(\mathbf{x}) - \rho_\varepsilon(\mathbf{x})), \quad \mathbf{x} \in \Omega, \quad (1.1)$$

where t denotes time, $\mathbf{x}$ the spatial variable, and θ a phenotypic variable. The unknown $n_\varepsilon(t, \mathbf{x}, \theta)$ is the structured population density, and

$$\rho_\varepsilon(t, \mathbf{x}) = \int_{\mathbb{T}} n_\varepsilon(t, \mathbf{x}, \theta) d\theta$$

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is the total population density. The function $K(\mathbf{x})$ represents the local carrying capacity, while the trait-dependent diffusivity $D(\theta)$ models the dispersal rate associated with phenotype θ . Throughout the paper, we assume that the dispersal rate is positive and periodic in the trait variable and that the carrying capacity is nonnegative and bounded:

$$0 < D_{\min} \leq D(\theta) \leq D_{\max} < \infty, \quad 0 \leq K(\mathbf{x}) \leq K_{\max}. \quad (1.2)$$

We impose homogeneous Neumann boundary conditions on $\partial\Omega$ and periodic boundary conditions in the θ -direction. Although our numerical development is carried out for (1.1), the main difficulty addressed here is more general and arises in a broader class of small-mutation trait-selection problems whose solutions become highly concentrated in the phenotypic direction.

Model (1.1) and related equations have been extensively studied from the analytical viewpoint, including well-posedness, long-time behavior, and the connection with adaptive dynamics [3, 4, 1]. A characteristic feature is that the competition between spatial heterogeneity and trait-dependent dispersal may select an optimal phenotype in the long-time regime. In many situations, the steady state exhibits concentration in the trait variable. The small parameter $\varepsilon > 0$ plays the role of a rare mutation scale, and in the limit $\varepsilon \rightarrow 0$ the steady state typically converges, in a weak sense, to a Dirac mass supported at the fittest trait. This singular limit is well understood at the PDE level through WKB or Hopf–Cole transforms and constrained Hamilton–Jacobi equations coupled with principal eigenvalue problems [4, 2, 3].

From the numerical point of view, however, this asymptotic regime remains challenging. As ε decreases, the trait profile of n_ε becomes increasingly narrow, so standard discretizations of the original variable require a rapidly growing number of grid points in the phenotypic direction. Moreover, approximating the full density with reasonable accuracy does not automatically guarantee an accurate identification of the dominant trait. This is particularly restrictive in the small- ε regime, where the quantity of real interest is often the location of the concentration and its effective profile rather than a pointwise resolution of the entire density on an extremely fine mesh.

Motivated by the asymptotic structure of the rare-mutation limit, we adopt the WKB representation

$$n_\varepsilon(\mathbf{x}, \theta, t) = W_\varepsilon(\mathbf{x}, \theta, t) \exp\left(\frac{u_\varepsilon(\theta, t)}{\varepsilon}\right),$$

where the phase variable u_ε describes the rapidly varying exponential profile in the trait direction, while W_ε is a comparatively smoother amplitude. In the long-time and rare-mutation regime, the effective behavior of the phase variable is related to a constrained Hamilton–Jacobi equation involving a principal eigenvalue problem in the spatial variable [4, 1]. This observation suggests that the WKB reformulation is not only asymptotically natural but also computationally advantageous. In particular, it provides a mechanism for extracting the dominant trait from the phase variable without directly resolving the narrow peak of the original density on a prohibitively fine trait grid.

The present paper is organized around this computational viewpoint. Our main goal is to develop and study a WKB-based numerical strategy that is able to capture trait concentration more efficiently than direct discretizations of the original model. To this end, we first derive a semi-discrete formulation for the phase and amplitude variables. We then specify a concrete numerical method based on a monotone discretization of the Hamilton–Jacobi part and a positivity-preserving discretization of the amplitude equation. At the analytical level, we establish positivity of the amplitude and derive a stationary, fixed-grid AP structure under transparent reconstruction and one-sided remainder stability assumptions. Since the WKB split does not automatically provide an exact discrete chain rule, we also record a density-compatible variant that restores a conservative density-level balance and makes the weighted remainder estimate verifiable. On the numerical side,

our experiments are designed to compare the original discretization and the WKB formulation directly, to track the remainder diagnostic, and to test the accurate capture of the fittest trait when ε is small.

The paper is organized as follows. In Section 2 we recall the WKB ansatz and the formal steady-state rare mutation limit that motivate the numerical formulation. In Section 3 we present the semi-discrete scheme, prove positivity, and derive the conditional stationary AP structure, including the density-compatible variant that controls the weighted remainder. Section 4 is devoted to numerical experiments, where we compare the WKB formulation with direct discretizations, examine its performance in the concentration regime, and monitor the remainder diagnostic. The appendices collect typical choices of monotone numerical Hamiltonians and monotone numerical fluxes, a fully discrete realization, and sufficient conditions for the one-sided remainder bound.

2 Preliminary

2.1 WKB ansatz and reformulated system

To capture the concentration phenomenon in the trait variable, we introduce the classical WKB ansatz

$$n_\varepsilon(x, \theta, t) = W_\varepsilon(x, \theta, t) \exp\left(\frac{u_\varepsilon(\theta, t)}{\varepsilon}\right). \quad (2.3)$$

This decomposition separates the exponentially varying part in the trait direction from a comparatively smoother amplitude. It is therefore natural for both the continuous asymptotic analysis and the numerical design in the small-mutation regime.

Substituting (2.3) into (1.1) and assigning the leading trait-direction exponential contributions to the phase function, we obtain the reformulated system

$$\begin{cases} \partial_t u_\varepsilon - |\partial_\theta u_\varepsilon|^2 - \varepsilon \partial_{\theta\theta} u_\varepsilon = -\mathcal{H}(\theta, t), \\ \varepsilon \partial_t W_\varepsilon - D(\theta) \Delta_x W_\varepsilon - \varepsilon^2 \partial_{\theta\theta} W_\varepsilon - 2\varepsilon \partial_\theta W_\varepsilon \partial_\theta u_\varepsilon = W_\varepsilon (K(x) - \rho_\varepsilon(x, t) + \mathcal{H}(\theta, t)), \end{cases} \quad (2.4)$$

where $\mathcal{H}(\theta, t)$ denotes an effective Hamiltonian. Using the identity

$$\partial_\theta (W_\varepsilon \partial_\theta u_\varepsilon) = \partial_\theta W_\varepsilon \partial_\theta u_\varepsilon + W_\varepsilon \partial_{\theta\theta} u_\varepsilon,$$

we rewrite the transport coupling in conservative form. If

$$F_\varepsilon = -W_\varepsilon \partial_\theta u_\varepsilon, \quad (2.5)$$

then

$$-2\varepsilon \partial_\theta W_\varepsilon \partial_\theta u_\varepsilon = 2\varepsilon \partial_\theta F_\varepsilon + 2\varepsilon W_\varepsilon \partial_{\theta\theta} u_\varepsilon.$$

Consequently, the system can be written as

$$\begin{cases} \partial_t u_\varepsilon - |\partial_\theta u_\varepsilon|^2 - \varepsilon \partial_{\theta\theta} u_\varepsilon = -\mathcal{H}(\theta, t), \\ \varepsilon \partial_t W_\varepsilon - D(\theta) \Delta_x W_\varepsilon - \varepsilon^2 \partial_{\theta\theta} W_\varepsilon + 2\varepsilon \partial_\theta F_\varepsilon = W_\varepsilon (K(x) - \rho_\varepsilon(x, t) + \mathcal{H}(\theta, t) - 2\varepsilon \partial_{\theta\theta} u_\varepsilon). \end{cases} \quad (2.6)$$

In the continuous theory, $\mathcal{H}$ is related to a principal eigenvalue problem in the spatial variable. This structure is the basic reason for introducing the variables $(u_\varepsilon, W_\varepsilon)$ in the first place.

2.2 Continuous asymptotic structure of the steady state

We next recall the steady-state asymptotic structure that motivates the numerical method. Let $(\bar{n}_\varepsilon, \bar{\rho}_\varepsilon)$ be a positive steady state of (1.1). Then

$$-D(\theta)\Delta_x \bar{n}_\varepsilon - \varepsilon^2 \partial_{\theta\theta} \bar{n}_\varepsilon = \bar{n}_\varepsilon (K(x) - \bar{\rho}_\varepsilon(x)), \quad \bar{\rho}_\varepsilon(x) = \int_{\mathbb{T}} \bar{n}_\varepsilon(x, \theta) d\theta. \quad (2.7)$$

As shown in [4], under standard assumptions on K , D , and the domain, the steady-state family admits several properties that are fundamental for the rare-mutation limit.

Proposition 2.1 (Continuous asymptotic structure of the steady state). *Assume that K is positive and nonconstant, that D is positive, periodic, and has a unique minimizer θ_m , and that the steady problem (2.7) admits a positive solution. Then the following properties hold at the continuous level.*

1. *There exists a constant $C > 0$, independent of ε , such that*

$$0 \leq \bar{\rho}_\varepsilon(x) \leq C \quad \text{for all } x \in \Omega. \quad (2.8)$$

Moreover, up to subsequences, $\bar{\rho}_\varepsilon$ converges uniformly to a limit density ρ .

2. *If we write*

$$\bar{n}_\varepsilon(x, \theta) = \exp\left(\frac{\bar{u}_\varepsilon(x, \theta)}{\varepsilon}\right), \quad (2.9)$$

then $\bar{u}_\varepsilon$ is uniformly Lipschitz continuous in θ , and along subsequences

$$\bar{u}_\varepsilon \rightarrow u \quad \text{uniformly in } (x, \theta), \quad (2.10)$$

where the limit u is independent of x , periodic in θ , and satisfies

$$\begin{cases} |\partial_\theta u(\theta)|^2 = \mathcal{H}(\theta, \rho(\cdot)), \\ \max_{\theta \in \mathbb{T}} u(\theta) = 0. \end{cases} \quad (2.11)$$

3. *The effective Hamiltonian $\mathcal{H}(\theta, \rho)$ is defined through the principal eigenvalue problem*

$$\begin{cases} -D(\theta)\Delta_x \mathcal{N}(x, \theta) = \mathcal{N}(x, \theta)(K(x) - \rho(x)) + \mathcal{N}(x, \theta) \mathcal{H}(\theta, \rho(\cdot)), & x \in \Omega, \\ \partial_\nu \mathcal{N} = 0, & x \in \partial\Omega, \end{cases} \quad (2.12)$$

with a positive eigenfunction $\mathcal{N}$. Moreover, $\mathcal{H}$ has the same monotonicity in θ as D , and hence

$$\min_{\theta \in \mathbb{T}} \mathcal{H}(\theta, \rho(\cdot)) = \mathcal{H}(\theta_m, \rho(\cdot)) = 0. \quad (2.13)$$

4. *The steady state concentrates on the fittest trait in the limit $\varepsilon \rightarrow 0$. More precisely,*

$$\bar{n}_\varepsilon \rightharpoonup N_m(x) \delta(\theta - \theta_m), \quad \bar{\rho}_\varepsilon \rightarrow N_m(x), \quad (2.14)$$

where N_m solves the reduced Fisher-type problem

$$\begin{cases} -D(\theta_m)\Delta_x N_m = N_m(K(x) - N_m), & x \in \Omega, \\ \partial_\nu N_m = 0, & x \in \partial\Omega. \end{cases} \quad (2.15)$$

Proposition 2.1 provides the continuous target of the numerical scheme. The boundedness of $\bar{\rho}_\varepsilon$ keeps the effective Hamiltonian in a bounded regime. The constrained Hamilton–Jacobi equation identifies the dominant trait through the maximizer of u . The concentration result (2.14) indicates that an asymptotic-preserving discretization should recover the reduced pair (N_m, θ_m) without resolving the original density on an extremely fine trait grid. This is the sense in which the WKB reformulation is used in the sequel.

3 WKB reformulated numerical scheme

In this section, we develop and analyze a semi-discrete numerical scheme for the WKB reformulation (2.6). The scheme is formulated in terms of the phase variable u_ε and the amplitude variable W_ε , with the total density n_ε reconstructed from the WKB representation.

3.1 Semi-discrete numerical discretization

We work at the semi-discrete level in order to separate the discretization in (x, θ) from the additional issue of time stepping. For clarity, we present the scheme in one spatial dimension with uniform grids in the x - and θ -directions. This choice is not essential for the WKB decomposition, but it keeps the notation transparent.

Let $\{x_j\}_{j=1}^J$ be a uniform grid on $\Omega = (a, b)$ and let $\{\theta_k\}_{k=1}^K$ be a uniform periodic grid on $\mathbb{T} = (0, 1)$, with mesh size $\Delta\theta$. We denote

$$W_{j,k}^\varepsilon(t) \approx W_\varepsilon(t, x_j, \theta_k), \quad u_k^\varepsilon(t) \approx u_\varepsilon(t, \theta_k).$$

For notational convenience, we write

$$D_k = D(\theta_k), \quad K_j = K(x_j).$$

By (1.2), we have

$$0 < D_{\min} \leq D_k \leq D_{\max} < \infty, \quad 0 \leq K_j \leq K_{\max}.$$

For a grid function $V = \{V_{j,k}\}$, we use the standard second-order difference operators

$$\delta_x^2 V_{j,k} = \frac{V_{j+1,k} - 2V_{j,k} + V_{j-1,k}}{(\Delta x)^2}, \quad \delta_\theta^2 V_{j,k} = \frac{V_{j,k+1} - 2V_{j,k} + V_{j,k-1}}{(\Delta\theta)^2}.$$

The homogeneous Neumann boundary condition in x is imposed by the ghost values

$$V_{0,k} = V_{1,k}, \quad V_{J+1,k} = V_{J,k},$$

while all indices in the θ -direction are understood periodically. For the phase variable, we also use

$$D_\theta^- u_k = \frac{u_k - u_{k-1}}{\Delta\theta}, \quad D_\theta^+ u_k = \frac{u_{k+1} - u_k}{\Delta\theta}.$$

The semi-discrete WKB system involves two numerical ingredients that are not contained in the standard second-order difference operators introduced above. The first one is the numerical Hamiltonian for the phase equation. The Hamilton–Jacobi term $-|\partial_\theta u_\varepsilon|^2$ is approximated by

$$\widehat{H}(D_\theta^- u_k^\varepsilon, D_\theta^+ u_k^\varepsilon).$$

We assume that $\widehat{H}$ is locally Lipschitz and consistent with the continuous Hamiltonian, namely

$$\widehat{H}(p, p) = -p^2.$$

We also assume that it is monotone in the sense that it is nondecreasing with respect to its first argument and nonincreasing with respect to its second argument. When $\widehat{H}$ is differentiable, this condition reads

$$\partial_a \widehat{H}(a, b) \geq 0, \quad \partial_b \widehat{H}(a, b) \leq 0.$$

Typical examples are the Godunov and Lax–Friedrichs numerical Hamiltonians, as recalled in Appendix A.

The second ingredient is the conservative discretization of the transport coupling in the amplitude equation. Recall that the continuous flux is

$$F_\varepsilon = -W_\varepsilon \partial_\theta u_\varepsilon.$$

We denote by

$$\mathcal{D}_\theta \widehat{\mathcal{F}}(W^\varepsilon, u^\varepsilon)_{j,k}$$

a conservative discretization of $\partial_\theta F_\varepsilon$ at (x_j, θ_k) . Here $\widehat{\mathcal{F}}$ approximates the flux $-W_\varepsilon \partial_\theta u_\varepsilon$. We assume that $\mathcal{D}_\theta \widehat{\mathcal{F}}$ is consistent with

$$\partial_\theta F_\varepsilon = \partial_\theta (-W_\varepsilon \partial_\theta u_\varepsilon)$$

when evaluated on smooth functions, and that it is conservative in the periodic trait variable. For the positivity argument below, we further assume the quasi-positivity property

$$W_{j,k} = 0, \quad W_{j,\ell} \geq 0 \text{ for all } \ell \implies -\mathcal{D}_\theta \widehat{\mathcal{F}}(W, u)_{j,k} \geq 0. \quad (3.16)$$

This property is satisfied by standard monotone finite volume fluxes, such as the upwind and Lax–Friedrichs fluxes listed in Appendix B.

With these two ingredients specified, we now define the discrete density and the semi-discrete WKB system. For notational convenience, set

$$E_k^\varepsilon(t) = \exp\left(\frac{u_k^\varepsilon(t)}{\varepsilon}\right). \quad (3.17)$$

The reconstructed nodal density is defined by

$$n_{j,k}^\varepsilon(t) = W_{j,k}^\varepsilon(t) E_k^\varepsilon(t). \quad (3.18)$$

As we shall see later, the evolution of u_ε and W_ε couples with the density function. Here we consider a general density computed by a trait-direction quadrature operator

$$\mathcal{Q}_\theta^\varepsilon[W_j^\varepsilon, u^\varepsilon],$$

which approximates

$$\int_{\mathbb{T}} n_\varepsilon(t, x_j, \theta) d\theta = \int_{\mathbb{T}} W_\varepsilon(t, x_j, \theta) \exp\left(\frac{u_\varepsilon(t, \theta)}{\varepsilon}\right) d\theta.$$

We define

$$\rho_{j,Q}^\varepsilon(t) = \mathcal{Q}_\theta^\varepsilon[W_{j,\cdot}^\varepsilon(t), u^\varepsilon(t)], \quad j = 1, \dots, J. \quad (3.19)$$

A direct composite quadrature gives

$$\rho_{j,Q}^\varepsilon(t) = m_j^\varepsilon(t) := \Delta\theta \sum_{k=1}^K W_{j,k}^\varepsilon(t) E_k^\varepsilon(t), \quad (3.20)$$

whereas in the small- ε regime one may instead use a Laplace-type reconstruction presented in Appendix C near the maximizer of u^ε .

Given

$$\boldsymbol{\rho}_Q(t) = (\rho_{1,Q}^\varepsilon(t), \dots, \rho_{J,Q}^\varepsilon(t)),$$

the discrete effective Hamiltonian $\mathcal{H}_k(\boldsymbol{\rho}_Q(t))$ is determined by the discrete principal eigenvalue problem

$$-D_k \delta_x^2 \mathcal{N}_j^{(k)} = \mathcal{N}_j^{(k)} (K_j - \rho_{j,Q}^\varepsilon(t)) + \mathcal{N}_j^{(k)} \mathcal{H}_k(\boldsymbol{\rho}_Q(t)), \quad j = 1, \dots, J. \quad (3.21)$$

Here $\mathcal{N}^{(k)}$ is the corresponding positive eigenvector, equipped with the same discrete Neumann boundary condition in x . Its normalization does not affect $\mathcal{H}_k$. For definiteness, one may impose

$$\Delta x \sum_{j=1}^J \mathcal{N}_j^{(k)} = 1.$$

With the above notation, the semi-discrete WKB scheme reads

$$\begin{cases} \frac{d}{dt} u_k^\varepsilon + \widehat{H} (D_\theta^- u_k^\varepsilon, D_\theta^+ u_k^\varepsilon) - \varepsilon \delta_\theta^2 u_k^\varepsilon = -\mathcal{H}_k(\boldsymbol{\rho}_Q(t)), & k = 1, \dots, K, \\ \varepsilon \frac{d}{dt} W_{j,k}^\varepsilon - D_k \delta_x^2 W_{j,k}^\varepsilon - \varepsilon^2 \delta_\theta^2 W_{j,k}^\varepsilon + 2\varepsilon \mathcal{D}_\theta \widehat{\mathcal{F}}(W^\varepsilon, u^\varepsilon)_{j,k} \\ = W_{j,k}^\varepsilon (K_j - \rho_{j,Q}^\varepsilon(t) + \mathcal{H}_k(\boldsymbol{\rho}_Q(t)) - 2\varepsilon \delta_\theta^2 u_k^\varepsilon), & j = 1, \dots, J, \quad k = 1, \dots, K. \end{cases} \quad (3.22)$$

We next show the equation satisfied by the reconstructed density $n_{j,k}^\varepsilon$. Since E_k^ε is independent of the spatial index j ,

$$\delta_x^2 n_{j,k}^\varepsilon = E_k^\varepsilon \delta_x^2 W_{j,k}^\varepsilon.$$

Moreover,

$$\varepsilon \frac{d}{dt} n_{j,k}^\varepsilon = E_k^\varepsilon \left(\varepsilon \frac{d}{dt} W_{j,k}^\varepsilon + W_{j,k}^\varepsilon \frac{d}{dt} u_k^\varepsilon \right). \quad (3.23)$$

Multiplying the first equation in (3.22) by $E_k^\varepsilon W_{j,k}^\varepsilon$ and multiplying the second equation in (3.22) by $\varepsilon E_k^\varepsilon$, noticing (3.23), we can derive that $n_{j,k}^\varepsilon$ satisfies

$$\varepsilon \frac{d}{dt} n_{j,k}^\varepsilon - D_k \delta_x^2 n_{j,k}^\varepsilon = (K_j - \rho_{j,Q}^\varepsilon) n_{j,k}^\varepsilon + R_{j,k}^\varepsilon, \quad (3.24)$$

where the remainder has the form

$$R_{j,k}^\varepsilon = \varepsilon^2 E_k^\varepsilon \delta_\theta^2 W_{j,k}^\varepsilon - \varepsilon W_{j,k}^\varepsilon E_k^\varepsilon \delta_\theta^2 u_k^\varepsilon - W_{j,k}^\varepsilon E_k^\varepsilon \widehat{H} (D_\theta^- u_k^\varepsilon, D_\theta^+ u_k^\varepsilon) - 2\varepsilon E_k^\varepsilon \mathcal{D}_\theta \widehat{\mathcal{F}}(W^\varepsilon, u^\varepsilon)_{j,k}. \quad (3.25)$$

Thus the reconstructed density satisfies a semi-discrete analogue of the original density equation, with $R_{j,k}^\varepsilon$ collecting the trait-direction discretization effects. When the numerical Hamiltonian and the conservative flux are evaluated on smooth exact functions, this remainder is consistent with the trait diffusion contribution $\varepsilon^2 \partial_{\theta\theta} n^\varepsilon$.

A density-compatible variant.

The split form above is convenient because it keeps the Hamilton–Jacobi and transport components explicit. For the stationary density estimate, however, it is useful to record an alternative discretization in which the reconstructed density satisfies a conservative density equation exactly. Keeping the same phase equation, replace the amplitude equation by

$$\begin{aligned} \varepsilon \frac{d}{dt} W_{j,k}^\varepsilon - D_k \delta_x^2 W_{j,k}^\varepsilon - \varepsilon^2 (E_k^\varepsilon)^{-1} \delta_\theta^2 (W_{j,\cdot}^\varepsilon E_\cdot^\varepsilon)_k \\ = W_{j,k}^\varepsilon (K_j - \rho_{j,Q}^\varepsilon + \mathcal{H}_k(\boldsymbol{\rho}_Q) + \widehat{H}(D_\theta^- u_k^\varepsilon, D_\theta^+ u_k^\varepsilon) - \varepsilon \delta_\theta^2 u_k^\varepsilon), \end{aligned} \quad (3.26)$$

where

$$\delta_\theta^2(W_{j,\cdot}^\varepsilon, E^\varepsilon)_k = \frac{W_{j,k+1}^\varepsilon E_{k+1}^\varepsilon - 2W_{j,k}^\varepsilon E_k^\varepsilon + W_{j,k-1}^\varepsilon E_{k-1}^\varepsilon}{(\Delta\theta)^2}.$$

Multiplying (3.26) by E_k^ε and using the phase equation gives

$$\varepsilon \frac{d}{dt} n_{j,k}^\varepsilon - D_k \delta_x^2 n_{j,k}^\varepsilon - \varepsilon^2 \delta_\theta^2 n_{j,k}^\varepsilon = (K_j - \rho_{j,Q}^\varepsilon) n_{j,k}^\varepsilon. \quad (3.27)$$

Thus, for this density-compatible variant, the remainder in (3.24) is exactly $R_{j,k}^\varepsilon = \varepsilon^2 \delta_\theta^2 n_{j,k}^\varepsilon$. Consequently, the weighted remainder can be controlled by periodic discrete summation by parts. This observation gives a concrete way to avoid the conditional remainder assumption used below; see Appendix D.

3.2 Basic structural property

The semi-discrete scheme (3.24) can be shown to be positive-preserving as long as the initial data is nonnegative. The detailed result is summarized as the following lemma.

Lemma 3.1 (Positivity of the amplitude). *Assume that $W_{j,k}^\varepsilon(0) \geq 0$ for all j, k . Then the semi-discrete amplitude equation in (3.22) preserves nonnegativity, namely*

$$W_{j,k}^\varepsilon(t) \geq 0 \quad \forall j, k, t \geq 0,$$

as long as the solution exists. Consequently,

$$n_{j,k}^\varepsilon(t) \geq 0 \quad \forall j, k, t \geq 0.$$

Proof. It is enough to verify the quasi-positivity of the right-hand side of the finite-dimensional ODE system for W^ε . From the amplitude equation in (3.22), we can write

$$\varepsilon \frac{d}{dt} W_{j,k}^\varepsilon = D_k \delta_x^2 W_{j,k}^\varepsilon + \varepsilon^2 \delta_\theta^2 W_{j,k}^\varepsilon - 2\varepsilon \mathcal{D}_\theta \widehat{\mathcal{F}}(W^\varepsilon, u^\varepsilon)_{j,k} + W_{j,k}^\varepsilon B_{j,k}(t),$$

where

$$B_{j,k}(t) = K_j - \rho_{j,Q}^\varepsilon(t) + \mathcal{H}_k(\rho_Q(t)) - 2\varepsilon \delta_\theta^2 u_k^\varepsilon(t).$$

Let $W^\varepsilon \geq 0$ and suppose that $W_{j,k}^\varepsilon = 0$ for some (j, k) . Then, for an interior spatial point,

$$\delta_x^2 W_{j,k}^\varepsilon = \frac{W_{j+1,k}^\varepsilon + W_{j-1,k}^\varepsilon}{(\Delta x)^2} \geq 0, \quad \delta_\theta^2 W_{j,k}^\varepsilon = \frac{W_{j,k+1}^\varepsilon + W_{j,k-1}^\varepsilon}{(\Delta \theta)^2} \geq 0.$$

The same conclusion holds at the boundary because of the Neumann ghost values in x -direction and the periodicity in θ -direction. The quasi-positivity assumption on the conservative flux operator implies that

$$-\mathcal{D}_\theta \widehat{\mathcal{F}}(W^\varepsilon, u^\varepsilon)_{j,k} \geq 0.$$

Combining all the inequalities, noticing the fact that $W_{j,k}^\varepsilon B_{j,k}(t) = 0$ since we assumed $W_{j,k}^\varepsilon = 0$, we finally have that, at this specific (j, k) , the following inequality holds,

$$\left. \frac{d}{dt} W_{j,k}^\varepsilon \right|_{W_{j,k}^\varepsilon=0} \geq 0.$$

Thus the vector field points inward on the boundary of the nonnegative cone. The standard invariance criterion for finite-dimensional ODE systems implies the claim. $\square$

3.3 Stationary AP structure under a one-sided stability condition

We now turn to stationary semi-discrete states. The goal is to identify which parts of the continuous rare-mutation structure are retained on a fixed grid. For the generic split WKB scheme, the only nonstandard point is the trait-direction remainder in the reconstructed density equation. We therefore state the stationary result under an explicit one-sided stability condition. This keeps the AP statement conditional and avoids claiming that the WKB remainder is automatically controlled for every trait discretization. A verifiable density-compatible variant is discussed in Appendix D.

By Lemma 3.1, stationary states obtained from nonnegative amplitudes satisfy

$$W_{j,k}^\varepsilon \geq 0, \quad n_{j,k}^\varepsilon = W_{j,k}^\varepsilon E_k^\varepsilon \geq 0.$$

All estimates below are for such nonnegative reconstructed densities.

Weighted density balance. Define

$$\eta_j^\varepsilon = \Delta\theta \sum_{k=1}^K \frac{n_{j,k}^\varepsilon}{D_k}.$$

Multiplying (3.24) by $1/D_k$ and summing over the trait variable gives the stationary weighted balance equation

$$-\delta_x^2 m_j^\varepsilon = \eta_j^\varepsilon (K_j - \rho_{j,Q}^\varepsilon) + \mathcal{R}_j^{\varepsilon,D}, \quad (3.28)$$

where

$$m_j^\varepsilon = \Delta\theta \sum_{k=1}^K n_{j,k}^\varepsilon, \quad \mathcal{R}_j^{\varepsilon,D} = \Delta\theta \sum_{k=1}^K \frac{R_{j,k}^\varepsilon}{D_k}.$$

The term $\mathcal{R}_j^{\varepsilon,D}$ is the discrete analogue of the continuous weighted mutation contribution

$$\varepsilon^2 \int_{\mathbb{T}} \frac{1}{D(\theta)} \partial_{\theta\theta} n^\varepsilon d\theta = \varepsilon^2 \int_{\mathbb{T}} n^\varepsilon \partial_{\theta\theta} \left(\frac{1}{D(\theta)} \right) d\theta.$$

At the continuous level this is bounded above by $C\varepsilon^2 \rho^\varepsilon$. For the split WKB discretization, however, the discrete operators act on the reconstructed density $n_{j,k}^\varepsilon = W_{j,k}^\varepsilon E_k^\varepsilon$, and an exact chain rule is not available. We isolate the needed property as follows.

Assumption 3.1 (One-sided weighted remainder stability). *There exist constants $C_R > 0$ and $r_h \geq 0$, independent of ε , such that every stationary state under consideration satisfies*

$$(\mathcal{R}_j^{\varepsilon,D})_+ \leq C_R m_j^\varepsilon + r_h, \quad j = 1, \dots, J. \quad (3.29)$$

This condition only controls the positive part of the weighted defect; a negative contribution is favorable in the maximum-principle estimate. It is a structural stability requirement on the trait discretization, not a consequence of the preceding consistency statement. For a direct density-level conservative discretization, and for the density-compatible WKB variant (3.26), the same type of estimate follows by discrete summation by parts. For the generic split WKB scheme, it should be checked analytically under additional structure or monitored numerically.

We also record the mild stability condition needed for the density reconstruction in the nonlocal reaction term.

Assumption 3.2 (Reconstruction stability). *There exist constants $c_Q, C_Q > 0$ and $e_Q \geq 0$, independent of ε , such that*

$$\rho_{j,Q}^\varepsilon \geq c_Q m_j^\varepsilon - e_Q, \quad j = 1, \dots, J, \quad (3.30)$$

$$\rho_{j,Q}^\varepsilon \leq C_Q m_j^\varepsilon + e_Q, \quad j = 1, \dots, J. \quad (3.31)$$

For the direct composite quadrature $\rho_{j,Q}^\varepsilon = m_j^\varepsilon$, Assumption 3.2 holds with $c_Q = C_Q = 1$ and $e_Q = 0$.

Proposition 3.1 (Conditional density bound for stationary semi-discrete states). *Let $(u^\varepsilon, W^\varepsilon)$ be a stationary semi-discrete state with nonnegative reconstructed density. Assume that Assumptions 3.1 and 3.2 hold. Then there exist constants $C_m, C_\rho > 0$, independent of ε , such that*

$$0 \leq m_j^\varepsilon \leq C_m, \quad 0 \leq \rho_{j,Q}^\varepsilon \leq C_\rho, \quad j = 1, \dots, J.$$

Proof. Since $n_{j,k}^\varepsilon \geq 0$,

$$\frac{1}{D_{\max}} m_j^\varepsilon \leq \eta_j^\varepsilon \leq \frac{1}{D_{\min}} m_j^\varepsilon.$$

Using (3.28) and (3.29), we obtain

$$-\delta_x^2 m_j^\varepsilon + \eta_j^\varepsilon \rho_{j,Q}^\varepsilon \leq K_{\max} \eta_j^\varepsilon + C_R m_j^\varepsilon + r_h.$$

Let j_* be a point where $M^\varepsilon := \max_j m_j^\varepsilon = m_{j_*}^\varepsilon$. Since $\delta_x^2 m_{j_*}^\varepsilon \leq 0$,

$$\eta_{j_*}^\varepsilon \rho_{j_*,Q}^\varepsilon \leq K_{\max} \eta_{j_*}^\varepsilon + C_R M^\varepsilon + r_h.$$

If $M^\varepsilon \leq e_Q/c_Q$, the bound is immediate. Otherwise, $\rho_{j_*,Q}^\varepsilon \geq c_Q M^\varepsilon - e_Q > 0$, and hence

$$\frac{M^\varepsilon}{D_{\max}} (c_Q M^\varepsilon - e_Q) \leq \left(\frac{K_{\max}}{D_{\min}} + C_R \right) M^\varepsilon + r_h.$$

This quadratic inequality gives a bound for M^ε independent of ε . The upper estimate for $\rho_{j,Q}^\varepsilon$ then follows from (3.31). $\square$

Stationary phase estimate. The density bound controls the coefficients in the discrete eigenvalue problem and hence the stationary phase equation.

Proposition 3.2 (Stationary phase estimate on a fixed trait grid). *Assume that*

$$0 \leq \rho_{j,Q}^\varepsilon \leq \bar{\rho}, \quad j = 1, \dots, J,$$

where $\bar{\rho}$ is independent of ε . Then the discrete effective Hamiltonian defined by (3.21) satisfies

$$|\mathcal{H}_k(\boldsymbol{\rho}_Q)| \leq C_{\mathcal{H}}, \quad k = 1, \dots, K,$$

where $C_{\mathcal{H}}$ is independent of ε .

Assume further that the stationary phase equation

$$\widehat{H}_G(D_\theta^- u_k^\varepsilon, D_\theta^+ u_k^\varepsilon) - \varepsilon \delta_\theta^2 u_k^\varepsilon = -\mathcal{H}_k(\boldsymbol{\rho}_Q) \quad (3.32)$$

holds with the Godunov numerical Hamiltonian

$$\widehat{H}_G(p^-, p^+) = -((p^-)_-)^2 - ((p^+)_+)^2.$$

Then

$$\Delta\theta \sum_{k=1}^K |D_\theta^+ u_k^\varepsilon|^2 \leq C_{\mathcal{H}}.$$

Consequently,

$$\max_k |D_\theta^+ u_k^\varepsilon| \leq \frac{\sqrt{C_{\mathcal{H}}}}{\sqrt{\Delta\theta}}.$$

If the phase is normalized by

$$\max_k u_k^\varepsilon = 0,$$

then

$$\max_k |u_k^\varepsilon| \leq \sqrt{C_{\mathcal{H}}}.$$

All constants are independent of ε , although they may depend on the fixed trait mesh.

Proof. The discrete eigenvalue problem is associated with the symmetric operator

$$-D_k \delta_x^2 - (K_j - \rho_{j,Q}^\varepsilon).$$

Its principal eigenvalue has the Rayleigh representation

$$\mathcal{H}_k(\rho_Q) = \min_{\varphi \neq 0} \frac{D_k \Delta x \sum_j |\delta_x^+ \varphi_j|^2 - \Delta x \sum_j (K_j - \rho_{j,Q}^\varepsilon) \varphi_j^2}{\Delta x \sum_j \varphi_j^2}.$$

Since $0 \leq K_j \leq K_{\max}$ and $0 \leq \rho_{j,Q}^\varepsilon \leq \bar{\rho}$, we have

$$\mathcal{H}_k(\rho_Q) \geq -K_{\max}.$$

Testing the quotient with a constant vector gives $\mathcal{H}_k(\rho_Q) \leq \bar{\rho}$. Thus

$$|\mathcal{H}_k(\rho_Q)| \leq \max\{K_{\max}, \bar{\rho}\} =: C_{\mathcal{H}}.$$

Set $q_k^\varepsilon = D_\theta^+ u_k^\varepsilon$. Then $D_\theta^- u_k^\varepsilon = q_{k-1}^\varepsilon$, and

$$-\hat{H}_G(D_\theta^- u_k^\varepsilon, D_\theta^+ u_k^\varepsilon) = ((q_{k-1}^\varepsilon)_-)^2 + ((q_k^\varepsilon)_+)^2.$$

Summing (3.32) over the periodic trait grid eliminates $\delta_\theta^2 u^\varepsilon$. Therefore

$$\begin{aligned} \Delta\theta \sum_{k=1}^K |q_k^\varepsilon|^2 &= -\Delta\theta \sum_{k=1}^K \hat{H}_G(D_\theta^- u_k^\varepsilon, D_\theta^+ u_k^\varepsilon) \\ &= \Delta\theta \sum_{k=1}^K \mathcal{H}_k(\rho_Q) \leq C_{\mathcal{H}}. \end{aligned}$$

The pointwise slope bound follows from the fixed-grid inequality $|q_k|^2 \leq (\Delta\theta)^{-1} \Delta\theta \sum_\ell |q_\ell|^2$. If $\max_k u_k^\varepsilon = 0$, connecting any node to a maximizer along the periodic grid gives

$$|u_k^\varepsilon| \leq \Delta\theta \sum_{\ell=1}^K |D_\theta^+ u_\ell^\varepsilon| \leq \left(\Delta\theta \sum_{\ell=1}^K |D_\theta^+ u_\ell^\varepsilon|^2 \right)^{1/2}.$$

This proves the claim. □

Formal AP trait-selection limit. We finally pass formally to the fixed-grid rare-mutation limit. The statement is deliberately fixed-grid and conditional: it identifies the limiting selection mechanism once the density and phase estimates above are available.

Proposition 3.3 (Formal AP trait-selection limit). *Assume that the stationary semi-discrete states satisfy Assumptions 3.1 and 3.2, and that the phase is normalized by*

$$\max_k u_k^\varepsilon = 0.$$

Assume also that the stationary phase estimate of Proposition 3.2 applies. Then, up to a subsequence on the fixed grids,

$$\rho_Q^\varepsilon \rightarrow \rho^0, \quad u^\varepsilon \rightarrow u^0.$$

Moreover, the limiting phase satisfies the discrete constrained Hamilton–Jacobi system

$$\widehat{H}(D_\theta^- u_k^0, D_\theta^+ u_k^0) = -\mathcal{H}_k(\rho^0), \quad \max_k u_k^0 = 0.$$

For the Godunov Hamiltonian $\widehat{H}_G$, every maximizer k_ of u^0 satisfies*

$$\mathcal{H}_{k_*}(\rho^0) = 0.$$

If $\mathcal{H}_k(\rho^0)$ has a unique zero at k_m , then $k_ = k_m$. Hence the selected grid trait is identified by*

$$\theta_{k_m} \in \arg \max_k u_k^0.$$

Proof. Proposition 3.1 and (3.31) give a uniform bound on ρ_Q^ε . Since the grids are fixed, a subsequence converges to some ρ^0 . The normalized phase is also uniformly bounded on the fixed trait grid by Proposition 3.2; hence, up to a further subsequence, $u^\varepsilon \rightarrow u^0$.

Passing to the limit in the stationary phase equation gives the discrete constrained Hamilton–Jacobi system, because $\varepsilon \delta_\theta^2 u_k^\varepsilon \rightarrow 0$ on the fixed grid. For the Godunov Hamiltonian, $\widehat{H}_G \leq 0$, and therefore $\mathcal{H}_k(\rho^0) = -\widehat{H}_G(D_\theta^- u_k^0, D_\theta^+ u_k^0) \geq 0$. At a maximizer k_* of u^0 ,

$$D_\theta^- u_{k_*}^0 \geq 0, \quad D_\theta^+ u_{k_*}^0 \leq 0,$$

so $\widehat{H}_G(D_\theta^- u_{k_*}^0, D_\theta^+ u_{k_*}^0) = 0$. Hence $\mathcal{H}_{k_*}(\rho^0) = 0$. If this zero is unique, the maximizer must be k_m . $\square$

The preceding results should be read as a stationary AP structure rather than a complete AP convergence theorem. The constants are uniform in ε for a fixed mesh, while a simultaneous mesh-refinement and rare-mutation limit would require additional estimates on the reconstruction, the phase slopes, and the WKB remainder.

4 Numerical experiments

This section specifies the numerical tests used to assess the WKB formulation. The experiments are designed around three questions: whether the dominant trait can be captured on a coarse trait grid, whether the method remains stable as ε decreases, and whether the one-sided remainder condition used in the stationary analysis is reasonable for the implemented discretization. The numerical values in the tables below are to be filled after the production runs.

4.1 Common computational setting

Unless otherwise stated, we use

$$\Omega = (0, 1), \quad \mathbb{T} = [0, 1),$$

with homogeneous Neumann boundary conditions in x and periodic boundary conditions in θ . The baseline spatial carrying capacity and dispersal landscape are

$$K(x) = K_0 + K_1 \cos(2\pi x), \quad K_0 = 1, \quad K_1 = 0.5, \quad (4.33)$$

$$D(\theta) = D_{\min} + D_1(1 - \cos(2\pi(\theta - \theta_m))), \quad D_{\min} = 0.2, \quad D_1 = 0.4, \quad \theta_m = 0.35. \quad (4.34)$$

Thus D has a unique minimum at θ_m . In the rare-mutation limit, this value is the expected selected trait for the continuous steady state.

The default initial data for the WKB variables are

$$u_k^0 = -\alpha_0(1 - \cos(2\pi(\theta_k - \theta_0))), \quad \alpha_0 = 0.05, \quad \theta_0 = 0.70, \quad (4.35)$$

$$W_{j,k}^0 = 1 + 0.1 \cos(2\pi x_j), \quad n_{j,k}^0 = W_{j,k}^0 \exp(u_k^0/\varepsilon).$$

The initial maximizer θ_0 is deliberately chosen away from θ_m so that the computations test trait selection rather than merely preservation of a well-prepared peak.

For the direct density computation we solve the original equation for $n_{j,k}$. For the WKB computation we solve (3.22); when explicitly stated, we also run the density-compatible variant (3.26). The discrete effective Hamiltonian is computed as the principal eigenvalue of

$$A_k(\rho_Q) = -D_k L_x - \text{diag}(K_j - \rho_{j,Q}),$$

where L_x is the Neumann finite-difference Laplacian. The phase is normalized after each time step by the gauge transformation

$$c^n = \max_k u_k^n, \quad u_k^n \leftarrow u_k^n - c^n, \quad W_{j,k}^n \leftarrow W_{j,k}^n \exp(c^n/\varepsilon),$$

which leaves $n_{j,k}^n = W_{j,k}^n \exp(u_k^n/\varepsilon)$ unchanged. Density reconstructions are evaluated in a log-stabilized form to avoid underflow:

$$\rho_{j,Q}^n = \exp(s^n/\varepsilon) \Delta\theta \sum_{k=1}^K W_{j,k}^n \exp\left(\frac{u_k^n - s^n}{\varepsilon}\right), \quad s^n = \max_k u_k^n.$$

The computation is stopped when the relative stationary residual satisfies

$$\mathcal{E}^n := \max \left\{ \frac{\|u^{n+1} - u^n\|_\infty}{\tau}, \frac{\|W^{n+1} - W^n\|_\infty}{\tau(1 + \|W^{n+1}\|_\infty)} \right\} \leq 10^{-9}. \quad (4.36)$$

For the direct density solver, the analogous residual is computed with $n^{n+1} - n^n$. In all tests the reported CPU time excludes plotting but includes the eigenvalue solves.

We use the following diagnostics. The trait marginal and selected traits are

$$P_k = \Delta x \sum_{j=1}^J n_{j,k}, \quad \theta_{\text{dir}} = \theta_{\arg \max_k P_k}, \quad \theta_{\text{WKB}} = \theta_{\arg \max_k u_k}.$$

The periodic distance on $\mathbb{T} = [0, 1)$ is

$$d_{\mathbb{T}}(a, b) = \min_{q \in \mathbb{Z}} |a - b + q|.$$

The concentration width of the reconstructed trait profile is measured by

$$\sigma_{\theta} = \left(\frac{\sum_{k=1}^K d_{\mathbb{T}}(\theta_k, \theta_{\text{WKB}})^2 P_k}{\sum_{k=1}^K P_k} \right)^{1/2}.$$

To monitor the assumption used in Proposition 3.1, we also record

$$\Gamma_R^{\varepsilon} = \max_{1 \leq j \leq J} \frac{(\mathcal{R}_j^{\varepsilon, D})_+}{m_j^{\varepsilon} + 10^{-14}}. \quad (4.37)$$

A bounded value of Γ_R^{ε} as ε decreases is not a proof of Assumption 3.1, but it is an important numerical check that the WKB remainder is not producing an uncontrolled positive contribution.

4.2 Test 1: direct density solver versus WKB solver

The first test compares the direct density discretization with the WKB solver on the baseline data (4.33)–(4.34). Use

$$J = 200, \quad K \in \{32, 64, 128\}, \quad \varepsilon \in \{5 \times 10^{-2}, 2 \times 10^{-2}, 10^{-2}\}.$$

A high-resolution direct computation with $K_{\text{ref}} = 1024$ is used as a finite- ε reference when feasible; otherwise the limiting trait θ_m is used as the reference selection target. The quantities to report are

$$e_{\text{dir}} = d_{\mathbb{T}}(\theta_{\text{dir}}, \theta_{\text{ref}}), \quad e_{\text{WKB}} = d_{\mathbb{T}}(\theta_{\text{WKB}}, \theta_{\text{ref}}).$$

Table 1: Test 1. Trait-selection accuracy of the direct and WKB solvers. Fill in the numerical values after the production runs.

ε	K	θ_{ref}	θ_{dir}	θ_{WKB}	CPU ratio
5×10^{-2}	32	—	—	—	—
5×10^{-2}	64	—	—	—	—
2×10^{-2}	32	—	—	—	—
2×10^{-2}	64	—	—	—	—
10^{-2}	64	—	—	—	—
10^{-2}	128	—	—	—	—

The expected outcome is that, for small ε , the direct method requires a much finer trait grid to place the peak correctly, whereas the WKB method identifies the selected trait through the smoother phase variable.

4.3 Test 2: fixed coarse trait grid and decreasing mutation scale

This test keeps a coarse trait mesh and decreases ε :

$$J = 200, \quad K = 64, \quad \varepsilon \in \{10^{-1}, 5 \times 10^{-2}, 2 \times 10^{-2}, 10^{-2}, 5 \times 10^{-3}\}.$$

The direct density solver is included until it becomes visibly under-resolved. For the WKB solver, report the selected trait, the concentration width σ_θ , and the minimum value of the discrete effective Hamiltonian at stationarity:

$$H_{\min} = \min_k \mathcal{H}_k(\rho_Q).$$

In the rare-mutation regime, θ_{WKB} should approach θ_m , while σ_θ should shrink with ε .

Table 2: Test 2. WKB behavior on a fixed coarse trait grid as ε decreases.

ε	θ_{WKB}	$d_{\mathbb{T}}(\theta_{\text{WKB}}, \theta_m)$	σ_θ	$H_{\min}$
10^{-1}	—	—	—	—
5×10^{-2}	—	—	—	—
2×10^{-2}	—	—	—	—
10^{-2}	—	—	—	—
5×10^{-3}	—	—	—	—

4.4 Test 3: mesh dependence in the trait direction

Here $\varepsilon = 10^{-2}$ is fixed and the trait grid is varied:

$$J = 200, \quad K \in \{16, 32, 64, 128, 256\}.$$

The aim is to show that the WKB prediction of the selected trait stabilizes on coarser grids than the direct density approximation. Plot P_k and u_k for representative values of K , and report the errors with respect to either the high-resolution reference or θ_m .

Table 3: Test 3. Trait-grid dependence for $\varepsilon = 10^{-2}$.

K	e_{dir}	e_{WKB}	direct CPU	WKB CPU
16	—	—	—	—
32	—	—	—	—
64	—	—	—	—
128	—	—	—	—
256	—	—	—	—

4.5 Test 4: nontrivial heterogeneous landscapes

To check that the method is not tuned to a single cosine profile, repeat the stationary computation with

$$K(x) = 1 + 0.35 \cos(2\pi x) + 0.20 \cos(4\pi x), \quad (4.38)$$

$$D(\theta) = 0.15 + 0.35(1 - \cos(2\pi(\theta - 0.25))) + 0.08(1 - \cos(4\pi(\theta - 0.25))). \quad (4.39)$$

This dispersal landscape still has its unique minimum at $\theta = 0.25$, but it has a sharper local shape near the selected trait. Use

$$J = 200, \quad K \in \{64, 128\}, \quad \varepsilon \in \{2 \times 10^{-2}, 10^{-2}\}.$$

The outputs are the total spatial density ρ_j , the trait marginal P_k , the phase u_k , and the selected trait. This test should be accompanied by plots comparing the direct and WKB reconstructions.

4.6 Test 5: remainder and density-compatibility diagnostics

The last test directly addresses the assumption used in the stationary analysis. For the baseline data, compute Γ_R^ε from (4.37) for the split WKB scheme and, separately, for the density-compatible variant (3.26). Use

$$J = 200, \quad K = 64, \quad \varepsilon \in \{5 \times 10^{-2}, 2 \times 10^{-2}, 10^{-2}, 5 \times 10^{-3}\}.$$

For the density-compatible variant, the theory in Appendix D predicts a bound controlled by the discrete second difference of $1/D_k$. For the split scheme, the table should be used as evidence of whether the one-sided defect remains moderate in the computed stationary states.

Table 4: Test 5. Numerical monitoring of the one-sided weighted remainder.

ε	Γ_R^ε , split WKB	Γ_R^ε , density-compatible WKB	selected trait
5×10^{-2}	—	—	—
2×10^{-2}	—	—	—
10^{-2}	—	—	—
5×10^{-3}	—	—	—

This diagnostic is important for presentation. If the split WKB remainder is observed to grow rapidly as ε decreases, then the numerical section should emphasize the density-compatible variant for stationary tests. If it remains bounded, the table provides practical support for the conditional stability assumption.

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A Typical choices of the monotone numerical Hamiltonian

We briefly list two choices of the numerical Hamiltonian used for the phase equation. For the continuous Hamiltonian

$$H(p) = -p^2,$$

the numerical Hamiltonian $\hat{H}(a, b)$ is required to satisfy $\hat{H}(p, p) = H(p)$, to be locally Lipschitz, and to be nondecreasing in its first argument and nonincreasing in its second argument.

A Lax–Friedrichs type choice is

$$\hat{H}^{\text{LF}}(a, b) = -\left(\frac{a+b}{2}\right)^2 + \frac{\alpha}{2}(a-b),$$

where α is chosen larger than the maximal slope magnitude reached in the computation.

The Godunov-type Hamiltonian used in the stationary phase estimate is

$$\hat{H}_G(a, b) = -(a_-)^2 - (b_+)^2, \quad a_- := \min\{a, 0\}, \quad b_+ := \max\{b, 0\}.$$

It is consistent since $\hat{H}_G(p, p) = -p^2$, and it has the required one-sided monotonicity.

B Typical realizations of the discrete conservative flux

We describe typical realizations of the abstract conservative operator

$$\mathcal{D}_\theta \widehat{\mathcal{F}}(W, u)_{j,k}$$

used in the split amplitude equation. Recall that the continuous flux is

$$F = -W \partial_\theta u.$$

On a uniform finite-volume grid in the trait variable, write

$$\mathcal{D}_\theta \widehat{\mathcal{F}}(W, u)_{j,k} = \frac{\widehat{F}_{j,k+\frac{1}{2}} - \widehat{F}_{j,k-\frac{1}{2}}}{\Delta\theta}.$$

Let

$$q_{k+\frac{1}{2}} = \frac{u_{k+1} - u_k}{\Delta\theta}, \quad a_{k+\frac{1}{2}} = -q_{k+\frac{1}{2}}.$$

Upwind flux. The standard upwind flux is

$$\widehat{F}_{j,k+\frac{1}{2}}^{\text{up}} = a_{k+\frac{1}{2}}^+ W_{j,k} + a_{k+\frac{1}{2}}^- W_{j,k+1},$$

where $a^+ = \max\{a, 0\}$ and $a^- = \min\{a, 0\}$. This flux is consistent with $-W \partial_\theta u$ and satisfies the quasi-positivity property used in Lemma 3.1.

Lax–Friedrichs flux. A Lax–Friedrichs flux is

$$\widehat{F}_{j,k+\frac{1}{2}}^{\text{LF}} = \frac{1}{2} a_{k+\frac{1}{2}} (W_{j,k} + W_{j,k+1}) - \frac{1}{2} \alpha_{k+\frac{1}{2}} (W_{j,k+1} - W_{j,k}),$$

with $\alpha_{k+\frac{1}{2}} \geq |a_{k+\frac{1}{2}}|$. Both the upwind and Lax–Friedrichs choices are conservative in the periodic trait variable:

$$\Delta\theta \sum_{k=1}^K \mathcal{D}_\theta \widehat{\mathcal{F}}(W, u)_{j,k} = 0.$$

C A fully discrete realization

We describe the fully discrete version used in the numerical tests. Assume that (u^n, W^n) is known at time t_n . First reconstruct the total density

$$\rho_j^n = \mathcal{Q}_\theta^\varepsilon[W_{j,\cdot}^n, u^n].$$

For the direct composite quadrature, use the log-stabilized formula

$$\rho_j^n = \exp(s^n/\varepsilon) \Delta\theta \sum_{k=1}^K W_{j,k}^n \exp\left(\frac{u_k^n - s^n}{\varepsilon}\right), \quad s^n = \max_k u_k^n.$$

A Laplace-type reconstruction may also be used as a diagnostic when u^n has a nondegenerate maximum θ_*^n :

$$\rho_j^n \approx W^n(x_j, \theta_*^n) \exp\left(\frac{u^n(\theta_*^n)}{\varepsilon}\right) \sqrt{\frac{2\pi\varepsilon}{|\partial_{\theta\theta} u^n(\theta_*^n)|}}.$$

For each trait node θ_k , compute H_k^n as the principal eigenvalue of

$$-D_k \delta_x^2 \mathcal{N}_j^{(k),n} = \mathcal{N}_j^{(k),n} (K_j - \rho_j^n) + \mathcal{N}_j^{(k),n} H_k^n, \quad j = 1, \dots, J,$$

with discrete Neumann boundary conditions. The eigenvector may be normalized by

$$\Delta x \sum_{j=1}^J \mathcal{N}_j^{(k),n} = 1.$$

The phase is updated by the semi-implicit scheme

$$\frac{u_k^{n+1} - u_k^n}{\tau} + \widehat{H} (D_\theta^- u_k^n, D_\theta^+ u_k^n) - \varepsilon \delta_\theta^2 u_k^{n+1} = -H_k^n.$$

For the split WKB scheme, the amplitude is updated by

$$\begin{aligned} \varepsilon \frac{W_{j,k}^{n+1} - W_{j,k}^n}{\tau} - D_k \delta_x^2 W_{j,k}^{n+1} - \varepsilon^2 \delta_\theta^2 W_{j,k}^{n+1} + 2\varepsilon \mathcal{D}_\theta \widehat{\mathcal{F}}(W^{n+1}, u^{n+1})_{j,k} \\ = W_{j,k}^{n+1} (K_j - \rho_j^n + H_k^n - 2\varepsilon \delta_\theta^2 u_k^{n+1}). \end{aligned} \quad (3.40)$$

The right-hand side uses ρ_j^n and H_k^n , so the update is linear in W^{n+1} once u^{n+1} is known. Other implicit–explicit variants are possible; the stationary tests only require convergence to the same semi-discrete equilibrium.

For the density-compatible variant, replace (3.40) by

$$\begin{aligned} \varepsilon \frac{W_{j,k}^{n+1} - W_{j,k}^n}{\tau} - D_k \delta_x^2 W_{j,k}^{n+1} - \varepsilon^2 (E_k^{n+1})^{-1} \delta_\theta^2 (W_{j,\cdot}^{n+1} E_k^{n+1})_k \\ = W_{j,k}^{n+1} (K_j - \rho_j^n + H_k^n + \widehat{H}(D_\theta^- u_k^{n+1}, D_\theta^+ u_k^{n+1}) - \varepsilon \delta_\theta^2 u_k^{n+1}), \end{aligned} \quad (3.41)$$

where $E_k^{n+1} = \exp(u_k^{n+1}/\varepsilon)$. In implementation the ratios $E_{k\pm 1}^{n+1}/E_k^{n+1}$ should be computed as exponentials of phase increments, not by forming very small exponentials separately.

After each phase update, apply the gauge normalization

$$c^{n+1} = \max_k u_k^{n+1}, \quad u_k^{n+1} \leftarrow u_k^{n+1} - c^{n+1}, \quad W_{j,k}^{n+1} \leftarrow W_{j,k}^{n+1} \exp(c^{n+1}/\varepsilon),$$

which preserves the reconstructed density.

D A sufficient condition for the one-sided remainder bound

We record two ways in which Assumption 3.1 can be justified or checked.

Density-compatible discretization.

For the density-compatible variant (3.26), the reconstructed density satisfies (3.27). Therefore, in the notation of (3.24),

$$R_{j,k}^\varepsilon = \varepsilon^2 \delta_\theta^2 n_{j,k}^\varepsilon.$$

Using periodic summation by parts in the trait variable,

$$\begin{aligned} \mathcal{R}_j^{\varepsilon,D} &= \varepsilon^2 \Delta \theta \sum_{k=1}^K \frac{\delta_\theta^2 n_{j,k}^\varepsilon}{D_k} \\ &= \varepsilon^2 \Delta \theta \sum_{k=1}^K n_{j,k}^\varepsilon \delta_\theta^2 \left(\frac{1}{D_k} \right). \end{aligned}$$

Hence, for $0 < \varepsilon \leq 1$,

$$(\mathcal{R}_j^{\varepsilon,D})_+ \leq C_{D,h} m_j^\varepsilon, \quad C_{D,h} := \max_k \left| \delta_\theta^2 \left(\frac{1}{D_k} \right) \right|.$$

This verifies (3.29) with $C_R = C_{D,h}$ and $r_h = 0$. If D is smooth and the second difference is the standard centered periodic difference, $C_{D,h}$ remains bounded along regular mesh refinement.

A sufficient condition for a generic split WKB discretization. For the split scheme (3.22), the same bound is not automatic because $R_{j,k}^\varepsilon$ contains discrete chain-rule defects. A simple sufficient condition is the following. Suppose that there exists $C_E > 0$, independent of ε , such that

$$\exp\left(\frac{u_{k+1}^\varepsilon - u_k^\varepsilon}{\varepsilon}\right) + \exp\left(\frac{u_{k-1}^\varepsilon - u_k^\varepsilon}{\varepsilon}\right) \leq C_E, \quad k = 1, \dots, K. \quad (4.42)$$

Assume also that the positive part of the local reconstruction defect satisfies

$$(R_{j,k}^\varepsilon)_+ \leq C_0 \sum_{\ell \in \{k-1, k, k+1\}} n_{j,\ell}^\varepsilon + r_{j,k,h}, \quad (4.43)$$

with C_0 independent of ε and

$$\Delta\theta \sum_{k=1}^K \frac{r_{j,k,h}}{D_k} \leq r_h.$$

Then positivity of n^ε , periodicity in k , and the lower bound on D_k give

$$(\mathcal{R}_j^{\varepsilon,D})_+ \leq C_R m_j^\varepsilon + r_h,$$

for a constant C_R independent of ε . Condition (4.42) is stronger than a uniform bound on the ordinary discrete derivative of u^ε . Indeed, $|D_\theta^+ u_k^\varepsilon| \leq C$ only gives $\exp(C\Delta\theta/\varepsilon)$, which is not uniform in ε on a fixed grid. This is why the main text treats (3.29) as a conditional stability assumption for the split WKB discretization.

E Discretization of the original density problem

For comparison, the original density equation is discretized as

$$\varepsilon \partial_t n_\varepsilon - D(\theta) \Delta_{xx} n_\varepsilon - \varepsilon^2 \Delta_\theta n_\varepsilon = n_\varepsilon (K(x) - \rho_\varepsilon(x)), \quad \rho_\varepsilon(x) = \int_0^1 n_\varepsilon(x, \theta) d\theta.$$

A typical semi-implicit time step is

$$\begin{aligned} \varepsilon \frac{n_{j,k}^{n+1} - n_{j,k}^n}{\tau} - D_k \delta_x^2 n_{j,k}^{n+1} - \varepsilon^2 \delta_\theta^2 n_{j,k}^{n+1} \\ = n_{j,k}^{n+1} (K_j - \rho_j^n), \quad \rho_j^n = \Delta\theta \sum_{k=1}^K n_{j,k}^n. \end{aligned}$$

The same spatial and trait grids are used for the direct and WKB solvers in all comparison tests.

References

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